Day 1 Checklist — Ready to Code

Nationwide TDP Capstone Project 2026 v2.0

Development Environment

▢ Windows 11 operational
▢ Visual Studio, Pycharm or VS Code working
▢ C# **or Python 3.14** development environment working
▢ .NET 8 SDK working where using C#
▢ Python **3.14** installed and working if using Python
▢ Git installed and configured
▢ Azure Functions Core Tools working
▢ Docker installed and working — optional

Repository & Collaboration

▢ GitHub account working
▢ Repository access confirmed
▢ Repository cloned successfully
▢ **main** and **dev** branches available
▢ Feature branch creation working
▢ Pull requests working
▢ Team communication channel available

Database & Data

▢ SQL environment accessible
▢ SQL connection working
▢ T-SQL queries can be executed
▢ Initial sample data available
▢ Customer spending data and emissions-factor data understood
▢ Basic data-quality considerations understood

Azure & Learning Platforms

▢ Team Azure account received — optional
▢ Azure login and subscription access confirmed — optional
▢ TeamQuest AI Learning Buddy access confirmed
▢ TeamQuest login working

CI/CD & Testing

▢ GitHub Actions pipeline accessible
▢ The build can be executed
▢ Unit testing framework working

▢ Initial test can be executed successfully
▢ Security/dependency checks available in the CI workflow

DevSecOps

▢ Security understood as part of development from the start
▢ No credentials or secrets stored in source code
▢ Input validation understood
▢ Secure SQL access understood
▢ Pull-request review process understood
▢ Basic security risks identified

AI-Assisted Development

▢ AI understood as a development assistant
▢ AI available for code explanation, debugging, SQL, testing and documentation
▢ AI-generated output will be reviewed and tested
▢ Confidential/customer information **will not** be entered into AI tools

Team & Professional Readiness

▢ Team members introduced and roles agreed
▢ Individual strengths identified
▢ Communication and escalation approach agreed
▢ Approach to problem-solving agreed – swarming?
▢ Collaboration across the different streams understood
▢ Technical, data, security and business perspectives recognised

Commercial & Customer Awareness

▢ Customer problem understood
▢ Purpose of the Carbon Footprint & Savings Advisor understood
▢ Importance of customer value understood
▢ **MVP requirements vs stretch features** understood
▢ Technical decisions will consider customer and commercial value

Sprint 1 Readiness

▢ Sprint 1 objective understood
▢ First MVP scope understood
▢ Expected customer outcome understood
▢ Expected technical outcome understood
▢ Expected professional outcomes understood

▢ Problem-solving aggreed
▢ Collaboration across disciplines
▢ Understanding and utilisation of individual strengths

▢ Communication
▢ Commercial awareness

Are you Ready-to-Code?

▢ Development environment works
▢ Repository works
▢ Database works
▢ CI/CD and testing work
▢ Security expectations understood
▢ AI-assistance approach understood
▢ Team roles and strengths understood
▢ Customer and commercial objectives understood
▢ Sprint 1 MVP understood

Then you are Ready to code, collaborate and deliver the first MVP.